OSF Amendment 1 — V1.11 (for submission)

Registration: Is a shared training recipe neutral? Recipe × architecture interactions in compact motor-imagery EEG decoders (DOI 10.17605/OSF.IO/8ST47, registered 1 Nov 2026)

Authors: Yu Gao, Yimin Zhang

Submitted: 13 November 2026 · **Status at submission:** neither confirmatory dataset (Lee2019-MI, Cho2017) has been downloaded, opened or analysed. No confirmatory run has been executed. All pilot work described here used only the two pilot datasets named in the registration (PhysioNet-MI, BCI IV 2a).

Co-author approval: Yimin Zhang has no OSF account. He reviewed and approved this amendment offline on 12 November 2026; his written approval is kept by the first author and available on request.

1. Purpose

This amendment (a) records the frozen confirmatory code, as the registration requires, (b) adds analyses and decision rules that were not specified in the original plan, and (c) records that these decision rules were fixed in a dated commit before the binary pilot of Section 3.3 was run, so that no pilot result could shape them. All additions are made before any confirmatory data are accessed and are therefore preregistered. Nothing in the original plan is removed; one statement is corrected (Section 4).

2. Frozen code and frozen decision rules

- Repository: github.com/jabarrick/recipe-x-arch
- Tag: v1.0-frozen · Commit: a44e5933ef73e311de86139a7ec1711fe3ed4497 (tagged 13 November 2026, 10:36 UTC; annotated tag object f6c3a37b8109d3bff748854d39c36b853453cea8)
- The pilot code remains on the csoanet-v2-deploy branch of github.com/jabarrick/CSOANet2026, which the new repository uses only as a dependency.
- Before tagging, the pipeline was run end-to-end on the pilot datasets only (PhysioNet-MI, BCI IV 2a) to check that it executes. No confirmatory dataset was touched.
- **Decision rules frozen before the binary pilot.** Every decision rule in this amendment was written into the file `docs/decision_rules.md` of the repository and committed on 8 November 2026, 09:09 UTC (commit f3fc1449bede7aabb5888b980d26ecf3912af59f, together with the analysis code and its unit tests), before the binary pilot of Section 3.3 was started. The file fixes: the B1 / B2 thresholds for H3 (Section 3.2); the home-recipe mapping, the equivalence bounds, the 80 % ranking criterion and the two estimation paths for incomplete tables (Section 3.1); the training-failure rule (Section 3.5, item 5); and the rule by which the binary pilot decides whether a third dataset is added (Section 3.3). The commit history of that file is the record that the rules preceded the pilot; the pilot results are disclosed in

supplementary table S7 and did not change any rule. Any later edit to the file is listed in the deviation table S4.

- **Both estimation paths of the home-recipe analysis are implemented and tested.** The subject-drop path and the sum-coded mixed-model path (Section 3.1) are both in the frozen code. Unit tests on synthetic 4 × 4 tables with known interaction residuals check that (i) the cell-mean formula recovers the planted residuals, (ii) the sum-coded model term agrees with the cell-mean formula on complete balanced tables to within numerical tolerance, and (iii) the two paths give the same answer when no cell is missing. The tests are part of the tagged commit.
- **Baseline implementations and reproduction tolerance.** EEGNet-8,2 is the PyTorch implementation in the CSOANet2026 repository (`eeg_project/scripts/models/eegnet.py`, kernels following Lawhern et al.'s fs/2 and fs/8 rule, max-norm 1 and 0.25); ShallowConvNet is implemented in this repository (`train/shallow.py`) and is numerically identical to braindecode 2.0 `ShallowFBCSPNet` (`tests/test_shallow.py`); MI-BMInet is our implementation (`deploy/mibminet.py` in the CSOANet2026 repository) following the paper and the authors' reference configurations. Before any confirmatory run, each baseline is trained with **its own published BCI IV 2a recipe** (not with R1–R4) on the pilot dataset BCI IV 2a and compared with one published figure, fixed here:
  - EEGNet-8,2: 60.46 % four-class accuracy on BNCI2014-001 (BCI IV 2a) in the MOABB benchmark (Chevallier et al., 2024; within-session evaluation, 5-fold cross-validation inside each session, MOABB `EEGNet_8_2` pipeline). Lawhern et al. (2018) report no numeric within-subject figure for this dataset in text or table (Figure 3 only, about 68 %, under a different protocol: 128 Hz, 4–40 Hz, 0.5–2.5 s); it is reported descriptively.
  - ShallowConvNet: 72.47 % four-class accuracy on BNCI2014-001 in the same MOABB benchmark (same protocol, `ShallowConvNet` pipeline). Schirrmeister et al. (2017), Table 2, give 73.7 % (0–38 Hz) and 71.9 % (4–38 Hz) with cropped training; these are reported descriptively because our training is trial-wise.
  - MI-BMInet: 76.03 % four-class accuracy on BCI IV 2a (Wang et al., 2024, Tables II–III), subject-specific, session 1 → session 2, 0–3 s after the cue, 0.5–100 Hz, N_k = 32, N_f = 64, Adam with constant learning rate 1e-3, 500 epochs, batch 32. The paper averages 25 repetitions; we use at least three seeds.
  - Tolerance: a reproduction is accepted if it lies within ±3 percentage points of the reference figure, with the reference protocol matched as closely as our pipeline allows; every remaining protocol difference is listed in S7 next to the two numbers. The tolerance was fixed before any baseline run. If a baseline falls outside the tolerance, the implementation is checked once against the reference and the check documented; if no error is found, the implementation is kept, the gap is reported in S7, and the home-recipe residual of that architecture (Section 3.1) is flagged as resting on an imperfect home recipe. The pilot value EEGNet under R2, within-subject (session split), BCI IV 2a = 67.1 % is a result of our pipeline, not a published figure, and is reported alongside.
  - Disclosure on R3: R3 (Adam 1e-2, ×0.1 at 40 % and 80 % of epochs, batch 16) is MI-BMInet's published **cross-subject** recipe (PhysioNet, 100 epochs, with N_k = 16 and N_f = 128 in that setting), not its BCI IV 2a within-subject recipe, which is

Adam 1e-3 constant, 500 epochs, batch 32. R3 is therefore MI-BMInet's home recipe for the cross-subject family F1 only; the home-recipe analysis is defined on F1, as registered, and the mapping is unchanged. The reference figures and this note were filled into `docs/decision_rules.md` on 8 November 2026 (they were marked [TO VERIFY] in commit f3fc1449; tolerance and handling rule unchanged); the edit is listed in S4.

- **Results (11 November 2026, three seeds each; S7).** EEGNet-8,2: 61.80 % against 60.46 % (+1.34, within tolerance). MI-BMInet: 74.05 % against 76.03 % (−1.98, within tolerance). ShallowConvNet: 66.62 % against 72.47 % (**−5.85, outside tolerance**). The pre-specified handling was applied: the implementation was checked against a reference implementation (it is numerically identical to braindecode 2.0 `ShallowFBCSPNet`; `tests/test_shallow.py`), no error was found, and the implementation is kept unchanged. The most likely source of the gap is a difference of model definition rather than of implementation: MOABB's Keras ShallowConvNet, which produced the reference figure, applies max-norm constraints (2.0 per filter on both convolutions, 0.5 on the classifier) that braindecode's and ours do not. A descriptive diagnostic with those constraints added (seed 0) gave 74.15 % (+1.68, within tolerance) against 66.37 % for the study definition with the same seed, with higher accuracy in all 18 subject × session units; the gap is therefore explained by the constraints, not by an implementation error. The study model is not changed: it follows braindecode's definition, and in the cross-subject binary pilot it was among the two best architectures under every recipe. The size of the difference on BCI IV 2a (about 8 points) is reported in S7 so that readers can weigh it. ShallowConvNet has no home recipe in the set, so the flag on the home-recipe residual does not apply; where ShallowConvNet's residual pattern is used as descriptive side-evidence (Section 3.1, rule 1), the gap is reported with it.

3. Additional preregistered analyses

3.1 Home-recipe analysis (secondary confirmatory)

Three architecture–recipe pairs in the design are "home" pairs: EEGNet with R2, MI-BMInet with R3 and CSOANet-v2 with R1. The coupling is not equally tight: CSOANet-v2 and R1 were developed together by the authors on the pilot data; R2 was published with EEGNet in 2018 on other data; R3 is MI-BMInet's published cross-subject recipe (used on PhysioNet with a different N_k / N_f), so it is a home recipe in the cross-subject family F1 but not in the within-subject family F2, where MI-BMInet's published recipe resembles R2. R4 (R1 without augmentation) is close to CSOANet-v2's home recipe. ShallowConvNet's published recipe is not in the set; it shares Adam and validation-based model selection with R2 but used a two-phase early-stopping scheme and cropped training, so it is only loosely "closest to R2".

H2b predicts that moving from R1 to R2 helps EEGNet more than CSOANet-v2. A co-adaptation account, in which each architecture performs best under the recipe it was developed with, predicts the same sign. H2b alone therefore cannot separate a robustness account from co-adaptation. Two indices are added to separate them.

The four-class pilot already shows why the separation must be made on interaction residuals rather than on simple differences. EEGNet improved from R1 to R2 (PhysioNet-MI four-class, cross-subject: 52.7 → 59.3) in the direction H2b predicts; but CSOANet-v2 also improved under R2 (same cell: 47.9 → 50.6; BCI IV 2a four-class within-subject: 66.9 → 69.6; subject-level means from the pilot result files `physionet_4class_summary.csv` and `bci2a_within_summary.csv` on the csoanet-v2-deploy branch). R2 may therefore be the stronger recipe overall. A "home minus away" difference would read CSOANet-v2's preference for R1 as negative and EEGNet's for R2 as inflated, purely because of the recipe main effect. These pilot numbers are disclosed as motivation and are not evidence for or against any hypothesis.

- **Home advantage** is defined from the cell means as the interaction residual of the home cell after removing both main effects: $h_a = \bar{y}_{a,home} - \bar{y}_{a,\cdot} - \bar{y}_{\cdot,home} + \bar{y}_{\cdot,\cdot}$. This formula is the definition used for all tests. In a balanced design it equals the $(\alpha\rho)_{a,home(a)}$ term of the registered model only when the model is fitted with sum-to-zero (effect) coding; with the default treatment coding the model's interaction coefficients are differences from a reference level and are not this quantity. The code computes h_a from cell means and, as a check, compares it with the sum-coded model term. A simple difference such as "home recipe minus the mean of the other recipes" is not used, because it also contains the recipe main effect: if R1 were better for every architecture, the simple difference would be positive for CSOANet-v2 regardless of any preference for its own recipe. The per-subject version $h[s,a]$ applies the same formula to each subject's 4 × 4 table (seed-averaged, float32, family F1).
  - Tests: two-sided one-sample t-test of $h[s,a]$ against 0 for each of EEGNet, MI-BMInet and CSOANet-v2, Holm-corrected at $\alpha = 0.05$. For CSOANet-v2 we additionally run an equivalence test (TOST, bounds ±2 points, 90 % CI). Absence of a home advantage is claimed only if equivalence is shown, never from a non-significant test.
  - Incomplete tables: the primary analysis has none, because training failures are kept as registered. In the sensitivity analysis that excludes training failures (Section 3.5, item 5), a subject whose 4 × 4 table loses any cell is dropped from the home-recipe analysis and the number dropped is reported. Because one cross-subject training failure removes a cell for every test subject of that fold, if more than 10 % of subjects would be dropped, h_a is instead estimated from the sum-coded mixed model fitted to all remaining cells. Both paths are implemented and tested in the frozen code (Section 2).
  - The primary analysis uses the full 4 × 4 table. Because R4 is close to CSOANet-v2's home recipe, a sensitivity analysis recomputes all residuals on the 4 × 3 table without R4 (the whole column is removed; removing R4 from one architecture only would break the double centring).
- **Recipe sensitivity**: for each architecture, the residual range (maximum minus minimum of its four interaction residuals) with a subject-level bootstrap 95 % CI (2,000 resamples). The raw range (best minus worst recipe mean) is also reported but is not used for decisions, because an architecture near ceiling or floor has a small raw range for reasons unrelated to recipe sensitivity. Because the residuals sum to zero over architectures within each recipe, they are coupled: a strong home effect of one

architecture pushes non-zero residuals onto the others in the same column. A small residual range therefore means that an architecture is, relative to this set of four, the closest to additive in architecture and recipe, not that it is insensitive to recipes in an absolute sense.

- **Decision rules**:
    1. CSOANet-v2's home residual is significantly positive (whatever the other architectures show): CSOANet-v2 prefers its own recipe beyond what the main effects explain. This is consistent with co-adaptation but cannot exclude a genuine fit between its structure and heavy augmentation; the two are indistinguishable in the home cell. H2b will not be interpreted as evidence of CSOANet-v2's robustness to the training recipe. As supporting, descriptive evidence only: if ShallowConvNet, which has no home recipe in the set, shows a residual pattern across recipes similar to CSOANet-v2's, this favours genuine fit over development history.

    2. CSOANet-v2's home residual is equivalent to zero and its residual range is the smallest of the four architectures in at least 80 % of bootstrap resamples: a robustness interpretation is supported, in the relative sense defined above. If EEGNet or MI-BMInet also show a significantly positive home residual, this is reported as the strongest form of that support.

       The thresholds are deliberately asymmetric: a single significant residual suffices for the interpretation that is unfavourable to the authors' own model, whereas the favourable interpretation requires both an equivalence test and a ranking criterion. We chose this because both CSOANet-v2 and R1 are the authors' own. The ranking criterion is set at 80 % rather than 95 % because strict ranking of four coupled residual ranges at 95 % would be reachable only with very large differences.

    3. Any other pattern: the mechanism behind H2b is reported as unresolved, with all effect sizes.

Fidelity of the home recipes. The registration fixes epochs and batch size per protocol, so the home recipes of EEGNet and MI-BMInet are approximations of their published recipes; CSOANet-v2's home recipe is unchanged. The changes are listed in a table in the manuscript and taken into account when interpreting the home-recipe analysis.

ShallowConvNet is reported descriptively as the architecture whose published recipe is closest to R2.

3.2 Outcome categories for H3

H3 is decided as registered. In addition, when H3 is not supported we will classify the 24 pairwise intervals (6 pairs × 4 recipes) as follows:

- **Stable (B1):** at least 16 of 24 intervals fall into a definitive category (A better, B better or equivalent) and no pair moves between two different definitive categories across recipes. Moves between inconclusive and a definitive category do not prevent B1, but are

counted and reported. Only this outcome will be described as evidence that a shared recipe preserves architecture conclusions.

- **Inconclusive (B2):** fewer than 8 of 24 intervals fall into a definitive category. This outcome will be described as uninformative about the neutrality of a shared recipe.
- Intermediate cases are reported as counts without an overall label.

The same 24 intervals and categories are computed on int8 accuracy, and the number of category changes from float32 to int8 is reported. This is a sensitivity analysis and decides no hypothesis.

We will also report, for H3, the number of category changes out of the 36 possible (6 pairs × 6 recipe pairs), separately for changes between two definitive categories (sign reversals or equivalent ↔ better) and changes between inconclusive and a definitive category.

Category precedence. A confidence interval can exclude zero and at the same time lie entirely within ±2 points (for example, 0.3 to 1.8). The registration does not say which category such an interval takes. It is classified as *A better* or *B better*; the *equivalent* label is used only for intervals that include zero and lie within ±2 points. This rule is fixed here and applied identically in S8 and in the confirmatory analysis.

Reading of equivalent ↔ better changes. With the precedence above, a true difference that is small but non-zero (between 0 and the SESOI) can be classified *better* under one recipe and *equivalent* under another by sampling variation alone; S8 shows that at the extended sample size this is the only source of false-positive H3 support under zero interaction (Section 3.3). We therefore fix the following reading in advance. H3 is decided as registered, counting any definitive ↔ definitive change. But if every definitive ↔ definitive change is of the equivalent ↔ better kind, with no sign reversal (*A better* ↔ *B better*), the result is reported as "pairwise differences below the SESOI whose direction is stable across recipes", and will not be described as a change in the ranking of architectures. The number of sign reversals is always reported next to the H3 decision, and S8 reports P(A) for the registered rule and for sign reversals only.

3.3 Binary pilot, expected interval width and outcome probabilities (attached file S8)

Binary pilot, run before this amendment and after the rules were frozen. The registered pilot was a four-class task, so its subject-level variability does not transfer directly to the two-class confirmatory design. We therefore ran a second pilot on the left- versus right-hand runs of PhysioNet-MI (runs 4, 8 and 12; subjects S088, S092, S100 and S104 excluded as in the registered pilot): all 4 architectures × 4 recipes, cross-subject 5-fold, one seed (seed 0), with the registered preprocessing and evaluation. Its only uses are (i) the subject-level balanced-accuracy table that feeds the simulation below and (ii) a check that the frozen pipeline runs end-to-end. It touches no confirmatory data. Its full results are disclosed in supplementary table S7 and are not evidence for or against any hypothesis; PhysioNet-MI is the dataset on which CSOANet-v2 and R1 were developed. The pilot was run on 9–10 November 2026 with the driver `train/pilot_binary.py` (commit 0eff2c8, after the rules were frozen at f3fc1449): 105 subjects, 44.8 scored trials per subject on average (42–45), 80 training runs, no training failures and no retries. Descriptively (S7), recipe R1 gave the lowest subject-level balanced accuracy for all four architectures (75.5–80.2 % against 79.4–81.8 % under R2), CSOANet-v2 was the lowest architecture under every recipe, the H2b contrast was +0.29 points (SD 10.83),

and all three home-cell interaction residuals were within ±0.2 points of zero. None of these numbers was used to change any rule.

Noise decomposition. A subject's accuracy in a pilot cell is estimated from about 45 trials, whereas the confirmatory cells have about 200 trials per subject (MOABB 1.7.2 documentation: Lee2019_MI loads the offline phase of both sessions, 100 labelled trials each; Cho2017 has 100 trials per class). The subject-level SD of a pairwise difference in the pilot is therefore inflated by binomial trial-sampling noise that will be much smaller in the confirmatory data. For each pair of architectures and each recipe, the observed variance of the subject-level difference d[s] is decomposed as Var_obs = Var_between + Var_binom, where Var_binom is the trial-sampling component computed from each subject's accuracies and trial count under a binomial model (for a difference between two cells, the sum of the two cells' binomial variances, i.e. assuming the two models' trial-level errors are uncorrelated within a subject; this is the conservative choice, since positively correlated errors would make the true component smaller). Var_between = max(Var_obs − Var_binom, 0) is the estimate of real between-subject variability; cases where the subtraction is truncated at zero are reported. The SD used for the confirmatory simulation is then recomposed at the confirmatory trial count: SD_conf = √(Var_between + Var_binom(n = 200)). Both the pilot SD and the recomposed SD are reported per pair in S8.

Scenario definitions (fixed here). *D* is the largest true pairwise architecture difference; architecture main effects are set to D × (0, 1/3, 2/3, 1), so the six pairwise differences are D/3, 2D/3 and D (for D = 3: 1, 2 and 3 points). *G* is the largest true interaction residual in absolute value; the pattern is a ±G swap between two architectures across two recipes, zero elsewhere, which is already double-centred. Between-subject covariance across the 16 cells is estimated from the pilot after removing the binomial component and projecting to positive semi-definite; if more than half of its eigenvalues are clipped, a simplified subject × architecture + residual model is used instead and this is stated in S8.

Simulation. With the recomposed covariance we simulate the expected half-width of the Bonferroni-adjusted 99.79 % intervals for each of the 6 architecture pairs at the planned sample size (n = 106; and n = 106 plus Dreyer2023 for the extended set), and the probabilities of outcome A (H3 supported), B1 and B2 over a grid of scenarios: D ∈ {1, 2, 3} points crossed with G ∈ {0, 1, 2, 3} points. P(A) under zero interaction is the family-wise false-positive rate of the registered H3 rule, given its 36 opportunities for a category change; it is reported explicitly, together with P(A) counting sign reversals only and the share of false positives that come from equivalent ↔ better changes (Section 3.2). Results (S8; script frozen at f3fc1449, default 2,000 simulations per scenario, seed 0). The full 16 × 16 between-subject covariance had 10 of 16 eigenvalues clipped to zero, so, as fixed above, the simplified subject × architecture + residual model was used; the full-covariance run is reported in S8 for comparison only. The binomial component was 7.9–8.2 points of the pilot SD of each pairwise difference (pilot SD 5.9–10.5 points); after recomposition at 200 trials the per-pair SD was 3.8–7.3 points, and the subtraction was truncated at zero in 5 of 24 pair × recipe cells. At n = 106 the median Bonferroni half-width is 1.78 points against the 2-point margin. Under zero interaction (G = 0) and D = 1, 2, 3: P(A) = 0.028, 0.035, 0.033 (all from equivalent ↔ better changes; P(A, sign reversals only) = 0); P(B1) = 0.001, 0.003, 0.052; P(B2) = 0.767, 0.614, 0.107. With an interaction of G = 2 points, P(A) ≥ 0.99 and P(A, sign reversals only) ≥ 0.97 for every D. At n = 193 (with Dreyer2023) the half-width

would be 1.31 points and P(B1) under G = 0 would be 0.28–0.48. Full grid and per-pair tables in S8.

Rule for adding a third dataset (fixed in `docs/decision_rules.md` before the pilot). The quantity that decides is P(B2) under the zero-interaction scenario with true pairwise differences of 1–3 points, at n = 106; this is the scenario in which H3 is not supported and the B1 / B2 distinction carries the result. If P(B2) > 0.5, Dreyer2023 is added as described below; otherwise it is not.

Decision (M1′, 10 November 2026). P(B2) under the deciding scenario was **0.496** (mean over D = 1, 2, 3 of 0.767, 0.614 and 0.107), which does not exceed 0.5, so **no third dataset is added** and H3 is kept as registered on Lee2019-MI and Cho2017. We record three facts about how robust this decision is. (i) The value lies 0.72 Monte Carlo standard errors (SE 0.0053) below the threshold; we did not rerun the simulation with other seeds or more replicates, because the frozen code's default run is the pre-specified one. (ii) The full-covariance model, which the frozen rule set aside because of eigenvalue clipping, would have given 0.608. (iii) The trial count used for recomposition (200 per subject) matches the MOABB documentation of both datasets, so the sensitivity run with 400 trials reported in S8 does not describe the confirmatory data. The consequence for interpretation is stated here in advance: at n = 106, outcome B1 has probability at most about 5 % even when there is no interaction, so a *stable* ranking is close to unreachable in this sample, and an *inconclusive* outcome (B2) has probability of about one half when true pairwise differences are 1–2 points. A B2 result will therefore be reported as uninformative about the neutrality of a shared recipe, as fixed in Section 3.2, and not as evidence for or against it.

3.4 Condition T: scoring seeds and a disclosed leakage

- **Scoring seeds.** Each of the 24 configurations in the condition-T search is scored as the mean validation accuracy over seeds 0 and 1. If the compute budget is exceeded (Section 3.7), a single seed (0) is used and this is reported.
- **T′, a leakage-free alternative, run before this amendment.** The same 24-configuration search with the same scoring rule (seeds 0 and 1) was run, before the code was frozen, on the left- versus right-hand motor-imagery runs of the pilot dataset PhysioNet-MI (runs 4, 8 and 12; subjects S088, S092, S100 and S104 excluded as in the pilot), cross-subject with the fold-1 tuning split defined in the registration. The configuration selected for each architecture is listed in Section 3.10 and is applied unchanged to all confirmatory cells. T′ therefore uses no confirmatory data and is a pre-specified recipe, not a procedure. Disclosure: PhysioNet-MI is the dataset on which CSOANet-v2 and R1 were developed, so the T′ search took place on the data the authors know best. T′ is reported alongside the fold-1-only analysis.
- **Correction to the registered text.** The registration states that test data are never used for the condition-T search. In the cross-subject cells this is not strictly true: the tuning split uses the training and validation subjects of fold 1, and those subjects are test subjects in folds 2–5. The search therefore uses data from subjects that are later tested, which favours condition T. The registered procedure is kept unchanged for the H4 decision. We add a sensitivity analysis that evaluates condition T on fold 1's test subjects only (about 11 subjects from Lee2019-MI and 10 from Cho2017, about 21 in total), where no such overlap exists. In the within-subject cell, the 10 tuning subjects are tested on

session 2, which was not used for tuning; their results are flagged in the per-subject output.

3.5 Additional sensitivity analyses

None of these decides a hypothesis. All are reported.

1. Random slopes: the F1 model refitted with by-subject random slopes for architecture and recipe (simplified to uncorrelated slopes if the full model fails to converge), and a subject-level cluster bootstrap (2,000 resamples). The bootstrapped statistics are: for H1, the variance share of recipe and the mean recipe range; for H2, the variance share of the architecture × recipe interaction; for H2b, the mean of the contrast c[s].
2. Fold: fold added as a fixed effect.
3. Dataset interactions: dataset × architecture, dataset × recipe and dataset × architecture × recipe added to the F1 model. If F1a and F1b analysed separately lead to opposite H2 or H3 conclusions, the pooled F1 conclusion will be reported as dataset-dependent.
4. Seeds: the model fitted to unaveraged accuracies with seed as an additional random effect. Per-cell seed SDs are reported for every cell (S3).
5. Training failures: a run is flagged as a training failure if its loss remains non-finite after the registered retry with seed + 100, or if its un-augmented training balanced accuracy at the selected checkpoint is ≤ 55 %. The rule uses training-side signals only; test accuracy is never used to flag or exclude a run. Primary analyses keep these runs as registered; a sensitivity analysis excludes them. If an H3 category change depends only on cells containing training failures, it will be reported as a recipe-robustness issue rather than an architecture × recipe interaction.
6. Model fit: likelihood-ratio tests on fixed effects use maximum-likelihood fits (not REML), as in the original plan.

3.6 Quantization reporting and calibration sample

For each architecture we report the share of operators quantized to int8 by ONNX Runtime static QDQ and list any that remain in float32. MinMax calibration is kept as registered and is described as the conservative choice.

Calibration sample. The registration fixes 256 training trials for calibration but not how they are chosen. They are drawn as follows and the procedure is written in S5: in the cross-subject cells, 256 trials sampled without replacement, uniformly over all training trials of that fold (training subjects only, never validation or test subjects), with a fixed random seed equal to the fold index; in the within-subject cell, 256 trials sampled the same way from that subject's session-1 training trials (seed = subject index), or all of them if fewer than 256 are available. The same calibration set is used for every architecture × recipe × seed within a fold or subject, so that differences in int8 accuracy between models cannot come from different calibration draws. Calibration trials are taken un-augmented. The indices of the sampled trials are saved with the exported models.

3.7 Compute budget and pre-specified cuts

The registered estimate of about 100 GPU-hours is revised to 150–200 GPU-hours. Runs are executed in this order: F1 under R1–R4; F2 under R1–R4; condition T for F1; T′ configurations for F1; condition T for F2; T′ configurations for F2. If the budget is exhausted, cuts are made in this order, and every cut is reported: 1. condition T for F2 (affects only H4 in the secondary family); 2. T′ configurations for F2; 3. scoring seeds in condition T for F1 reduced from 2 to 1.

No cut touches the runs that decide H1, H2, H2b or H3.

3.8 Smallest effect size of interest

The registered SESOI of 2 percentage points is retained. Rationale (added for transparency, not a change): 2 points is a substantive margin, not one derived from power. It is of the same order as the seed-to-seed variability of a single training configuration in our own pilot work (median absolute difference between seeds 0 and 1 of the same configuration: 1.1–1.3 points for EEGNet, ShallowConvNet and MI-BMInet, 2.1 for CSOANet-v2; T′ search, S1), so a difference within ±2 points is one that a change of seed alone could produce or erase. The expected standard error of a seed-averaged cell mean (about 0.6–1 point) is reported separately as a detection floor, not as a reason for the SESOI.

3.9 Non-substantive fixes after data download

Some problems in data loading can only appear once the confirmatory datasets are downloaded (for example, session structure, event codes or channel names). Changes made after download are allowed without counting as deviations only if they are confined to data loading and input/output code and do not touch the models, recipes, splits, seeds or analysis code. Each such fix is a separate commit and is listed in supplementary table S4 with its hash and reason. Any other change after download is reported as a deviation.

3.10 T′ configurations (selected before the code was frozen)

Search run on 11 November 2026 with `train/tprime_search.py` (commit 531c7fa), over the 24 configurations fixed in Supplementary S1 before the search (S4 row 3); the same 24 configurations for every architecture; PhysioNet-MI left vs right, fold 1 of the pilot split, scored on fold 1's validation subjects as the mean of seeds 0 and 1 (192 runs, none non-finite). Score = validation balanced accuracy (%).

Architecture	Config	Optimiser	Learning rate	Schedule	Batch	Augmentations	Label smoothing	Selection rule	Score	Runner-up
EEGNet	#10	Adam	3.06e-03	×0.1 at 40%/80%	64	noise, shift, channel dropout	0.0	all epochs, lowest val loss	86.41	85.86
Shallo	#1	Adam	3.40e-	cosine	32	none	0.0	early	88.65	88.51

Architecture	Config	Optimiser	Learning rate	Schedule	Batch	Augmentations	Label smoothing	Selection rule	Score	Runner-up
wConvNet			04	to 0.01×lr				stop on val bacc (patience 15)		
MI-BMInet	#13	AdamW (wd 1e-3)	2.53e-03	×0.1 at 40%/80%	32	noise	0.0	early stop on val bacc (patience 15)	89.77	89.64
CSOANet-v2	#10	Adam	3.06e-03	×0.1 at 40%/80%	64	noise, shift, channel dropout	0.0	all epochs, lowest val loss	87.51	85.29

The runner-up was within 0.1–0.6 points for EEGNet, ShallowConvNet and MI-BMInet, smaller than the typical seed-to-seed difference of a configuration's score (median 1.1–1.3 points), so the choice among the top configurations is largely noise, as anticipated in Section 3.4. These configurations are applied unchanged to every confirmatory cell.

4. Summary of changes

Item	Original plan	This amendment
Frozen code	to be recorded	tag v1.0-frozen, commit a44e5933
Decision rules	—	all rules fixed in `docs/decision_rules.md`, dated commit before the binary pilot; pilot results in S7
Home-recipe analysis	—	added, secondary confirmatory: home-cell interaction residual from cell means (equivalence test for CSOANet-v2), residual range; asymmetric thresholds stated; both estimation paths implemented and unit-tested
H3 outcome categories	not specified	B1 / B2 thresholds; change counts; same categories on int8; category precedence (better over equivalent) fixed; equivalent ↔ better-only support read as "below SESOI, stable direction", not a ranking change
Binary pilot	four-class	PhysioNet-MI left/right, 4 × 4, 5-fold, 1 seed, run after the rules

Item	Original plan	This amendment
	pilot only	were frozen; feeds S8 only
Interval-width simulation	—	S8 attached; subject-level SD decomposed into between-subject and binomial components and recomposed at the confirmatory trial count; scenario parameters D and G defined; P(A) reported for the registered rule and for sign reversals only
Third dataset	two datasets	decided by P(B2) > 0.5 under the zero-interaction scenario; P(B2) = 0.496, so no dataset added; the low reachability of B1 at n = 106 is stated in advance
Baseline implementations	not specified	each compared with one named figure within ±3 points: EEGNet +1.34, MI-BMInet −1.98 (within); ShallowConvNet −5.85 (outside; implementation verified and kept, gap reported, diagnostic in S7)
Condition-T scoring seeds	not specified	2 seeds (1 if budget cut)
Condition-T test-data statement	"never used"	corrected; fold-1-only analysis added
T′	—	search over the S1 space run on PhysioNet-MI before freezing (11 Nov 2026); selected configurations listed (3.10)
Training-failure rule	non-finite loss, retry once	plus training accuracy ≤ 55 %; no test-based rule
Fixes after download	not specified	allowed for data-loading code only, each listed
Sensitivity analyses	int8, ART-ANOVA, F1a/F1b	plus random slopes / bootstrap, fold, dataset interactions, seed random effect, training-failure exclusion
Quantization reporting	int8 accuracy, agreement	plus operator coverage
int8 calibration sample	256 training trials	fixed-seed draw from the fold's training subjects (or the subject's session-1 training trials); same set for all models in a fold; indices saved (S5)
Compute	about 100 GPU-hours	150–200 GPU-hours; ordered cuts

5. Timing

- Decision rules committed: 8 November 2026 (commit f3fc1449)

- Binary pilot run: 9–10 November 2026 (driver commit 0eff2c8); S8 and decision M1′: 10 November 2026
- Code frozen: tag v1.0-frozen, 13 November 2026
- Amendment submitted: 13 November 2026
- Confirmatory data first downloaded: only after this update is public on the registration (expected 7 Dec 2026)

Attachments

- S8 (folder `docs/S8/` of the repository: decision file, scenario grid, per-pair SD decomposition, half-widths, figure, comparison and sensitivity runs)
- decision_rules.md as of commit f3fc1449 (copy of the frozen rules file; later edits to the file are limited to its own change log and are listed in S4)